

to their root cause in the analogue waveform. Finally, oscilloscopes provide easy connectivity to serial buses through a variety of probing options and fixtures. There is no need for a standard port or special input/output port for connectivity."

Better graphical user interface

Protocol analysers with graphical user interfaces (GUIs) are gaining popularity among engineers due to their ease of use. The improved GUIs allow the users to execute test setups very quickly, resulting in improved productivity.

"User interfaces for these tools are becoming more and more user-friendly in the way they display decoded data, and hence protocol intricacies can now be dealt easily even with basic knowledge of protocol. It makes the task of test engineers fast and easy, while still maintaining the accuracy of testing. Some tools also support automation scripts for testing and analysis, which further helps reduce efforts and improve accuracy," explains Siddharth Patel, lead engineer-embedded software, EATON India Engineering Centre.

More intuitive GUIs also have the advantage of reducing the learning curve for an engineer, translating to more productivity on the whole.

"The embedded engineers are often required to monitor analogue and digital signals and some protocol analysis. Serial protocol analysis on digital channels enables viewing of analogue, digital and protocol analysis. To top it all, RTO's intuitive user interface helps engineers to observe all these signals easily with drag and drop, gestures, etc," explains Srinivasa Appalla, area manager, product support & applications, Test & Measurement, Rohde & Schwarz India.

Improved hardware

Design engineers do not want to be impacted by slow tools. Rather, they need tools that will move with their development speed as they iteratively

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test and debug their systems.

"One of the most valuable features of a protocol analyser is its ability to provide real-time feedback. Real-time feedback allows developers to capture, display and filter data instantly instead of waiting for the capture to download. With the data shown immediately on the screen and the engineer ready to perform interactive analysis with the captured data, the engineer's efficiency level can increase as a tool that matches their development pace is utilised," explains Fung.

Removing sluggishness while decoding helps improve the quality of work as well. Siddharth adds, "Instead of just software-based analysers, we now see more and more hardware-based protocol analysers with sophisticated PC user interface. Hardware analysers help in getting real-time stamp for frames and hence have very realistic picture of what is happening on the bus and when it is happening. With increasing buffer space on hardware and fast communication interface with PC software, these tools can capture a good amount of traffic continuously even with loaded bus."

Hardware-based decoding features are also available on scopes based on protocol analysers and help them overcome speed limitations.

"Hardware-based decoding provides a virtual real-time update of the decode trace, and doesn't degrade the scope's waveform update rate (which can be up to 1,000,000 waveforms per second). This enhances the scope's probability of capturing and display-

ing infrequent serial bus communication errors," says Bhatia.

In addition to improving the computational throughput of CPUs in software-defined test systems, Moore's law is driving enhancements in another type of computing device: the field-programmable gate array (FPGA). Essentially, FPGAs are pieces of hardware that you can reconfigure with software. These enable the engineers to reduce test time and perform tests not previously possible without custom hardware.

Satish Mohanram, technical marketing manager, National Instruments India, explains the benefits of FPGA-based encode and decode: "Instead of using software to encode signal information into a protocol-level signal (and vice versa), FPGAs can encode and decode data directly in hardware. This simplifies test system software because it needs to work with only data information, and it makes protocol-level communication possible where predefined test vectors alone are not suitable (for example, when quick decisions are needed based on protocol data). It might also be necessary to quickly transition from receiving to transmitting data on the same line. The test hardware must be able to detect and respond to this case."

In a nutshell

Enhancing the productivity and the accuracy are important goals for any design house. Protocol analysers are well positioned to help you achieve these goals. ●

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